

Method of Variation of Parameters

The second order differential equation is solved by this method.

The eqⁿ of the form,

$$\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = R$$

$$C.F. = C_1 u + C_2 v$$

Now,

$$\text{Wronskian (W) of } u \text{ \& } v = \begin{vmatrix} u & v \\ u' & v' \end{vmatrix}$$

where, u' & v' are the first derivative of ~~the~~ u & v .
~~general solution~~

The general solution of the given differential equation is,

$$y = C.F. + P.I. = C_1 u + C_2 v + u.f(x) + v.g(x)$$

$$P.I. = u f(x) + v g(x)$$

$$\text{where, } f(x) = - \int \frac{v R}{W} dx$$

$$\& \quad g(x) = \int \frac{u R}{W} dx$$

Solve:

Solve the differential solution by applying method of variation of parameters.

1. $y'' + n^2 y = \sec nx$

Solⁿ:

Let, the solution of the given differential equation be,

$$y = C.F. + P.I. = C_1 u + C_2 v + u f(x) + v g(x)$$

The given eqⁿ is,

$$y'' + n^2 y = \sec(xn) \\ \Rightarrow (D^2 + n^2)y = \sec(xn)$$

The A.E. is,

$$m^2 + n^2 = 0 \\ \Rightarrow m = \pm ni$$

$$C.F. = C_1 \underbrace{\cos nx}_u + C_2 \underbrace{\sin nx}_v$$

Now,

$$Wronskian(W) = \begin{vmatrix} \cos nx & \sin nx \\ -n \sin nx & n \cos nx \end{vmatrix} = n$$

$$f(x) = - \int \frac{\sin nx \cdot \sec nx}{n} dx = \frac{\log(\cos nx)}{n^2}$$

$$\& g(x) = \int \frac{\cos nx \cdot \sec nx}{n} dx = \frac{x}{n}$$

$$P.I. = uf(x) + vg(x)$$

$$= \frac{1}{n^2} \cos nx \cdot \log(\cos nx) + \frac{x \sin nx}{n}$$

Hence, the complete solution is,

$$y = C_1 \cos nx + C_2 \sin nx + \frac{1}{n^2} \cos nx \cdot \log(\cos nx) + \frac{x}{n} \sin nx$$

$$2. \quad y^2 + a^2 y = \tan ax$$

Solⁿ:

$$[D^2 + a^2]y = \tan ax$$

The A.E. is,

$$m^2 + a^2 = 0$$

$$\Rightarrow m = \pm ai$$

$$C.F. = C_1 \underbrace{\cos ax}_u + C_2 \underbrace{\sin ax}_v$$

Now,

$$W = \begin{vmatrix} \cos ax & \sin ax \\ -a \sin ax & a \cos ax \end{vmatrix} = a$$

$$P.I. = u f(x) + v g(x)$$

$$f(x) = - \int \frac{\sin ax \cdot \tan ax}{a} dx = \frac{1}{a^2} \sin ax - \frac{1}{a^2} \log(\sec ax + \tan ax)$$

$$\& g(x) = \int \frac{\cos ax \cdot \tan ax}{a} dx = -\frac{1}{a^2} \cos ax$$

$$P.I. = \cancel{\frac{1}{a^2} \cos ax \cdot \sin ax} - \frac{1}{a^2} \cos ax \cdot \log(\sec ax + \tan ax) \\ - \cancel{\frac{1}{a^2} \sin ax \cdot \cos ax}$$

$$\therefore y = C_1 \cos ax + C_2 \sin ax - \frac{1}{a^2} \cos ax \cdot \log(\sec ax + \tan ax)$$

$$3. y'' + a^2 y = \operatorname{cosec} ax$$

Solⁿ:

$$(D^2 + a^2)y = \operatorname{cosec} ax$$

The A.E. is,

$$m^2 + a^2 = 0$$

$$\Rightarrow m = \pm ai$$

$$C.F. = C_1 \underbrace{\cos ax}_u + C_2 \underbrace{\sin ax}_v$$

Now,

$$W = \begin{vmatrix} \cos ax & \sin ax \\ -a \sin ax & a \cos ax \end{vmatrix} = a$$

$$f(x) = - \int \frac{\sin ax \cdot \operatorname{cosec} ax}{a} dx = -\frac{x}{a}$$

$$g(x) = \int \frac{\cos ax \cdot \operatorname{cosec} ax}{a} dx = \frac{1}{a^2} \log(\sin ax)$$

$$P.I. = \frac{-x}{a} \cos ax + \frac{1}{a^2} \sin ax \cdot \log(\sin ax)$$

$$\therefore y = C.F. + P.I.$$

$$= C_1 \cos ax + C_2 \sin ax - \frac{x}{a} \cos ax$$

$$+ \frac{1}{a^2} \sin ax \cdot \log(\sin ax)$$

4. $y_2 - 3y_1 + 2y = \frac{e^x}{1+e^x}$

Solⁿ:

$$[D^2 - 3D + 2]y = \frac{x}{1+e^x}$$

The A.E. is,

$$\begin{aligned} m^2 - 3m + 2 &= 0 \\ \Rightarrow m^2 - (2+1)m + 2 &= 0 \\ \Rightarrow m^2 - 2m - m + 2 &= 0 \\ \Rightarrow m(m-2) - (m-2) &= 0 \\ \Rightarrow (m-2)(m-1) &= 0 \\ \Rightarrow m &= 1, 2 \end{aligned}$$

$$C.F. = C_1 \frac{e^x}{u} + C_2 \frac{e^{2x}}{v}$$

$$\text{Now, } W = \begin{vmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{vmatrix} = e^{3x}$$

$$f(x) = - \int \frac{e^{2x} \cdot e^x}{e^{3x}(1+e^x)} dx$$

$$= - \int \frac{1}{1+e^x} \cdot dx$$

$$= - \int \frac{e^{-x}}{e^{-x}(1+e^x)} dx$$

$$= - \int \frac{e^{-x}}{e^{-x} + 1} \cdot dx$$

$$\text{let, } t = e^{-x} + 1$$

$$\Rightarrow dt = -e^{-x} dx$$

$$f(x) = \int \frac{1}{t} dt$$

$$= \log(e^{-x} + 1) \quad [\text{or, } \log(1 + e^x) - x] = f(x)$$

$$g(x) = \int \frac{e^x \cdot e^x}{e^{3x}(1 + e^x)} \cdot dx$$

$$= \int \frac{1}{e^x(1 + e^x)} \cdot dx$$

$$= \int \frac{e^{-x}}{1 + e^x} \cdot dx$$

$$= \int \frac{e^{-x} \cdot e^{-x}}{e^{-x}(1 + e^x)} \cdot dx$$

$$= \int \frac{e^{-2x} + e^{-x} - e^{-x}}{e^{-x} + 1} \cdot dx$$

$$= \int \frac{e^{-x}(e^{-x} + 1) - e^{-x}}{e^{-x} + 1} \cdot dx$$

$$= \int e^{-x} dx - \int \frac{e^{-x}}{e^{-x}+1} dx$$

$$= -e^{-x} + \log(e^{-x}+1) \quad [\text{or } f(x) = -e^{-x} + \log(e^{-x}+1) - x]$$

$$\text{So, P.I.} = e^x \cdot \log(e^{-x}+1) - e^{-x} \cdot e^{2x} + e^{2x} \cdot \log(e^{-x}+1)$$

$$= e^x \cdot \log(e^{-x}+1) - e^x + e^{2x} \cdot \log(e^{-x}+1)$$

Hence, the complete solution is,

$$y = C.F. + P.I.$$

$$= C_1 e^x + C_2 e^{2x} + e^x \log(e^{-x}+1) - e^x + e^{2x} \log(e^{-x}+1)$$